

Data-AI-Experience (DAX) Challenge 2026

Technical Report

MERIDIAN: Strategic Underemployment & Pahang Vulnerability

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1.0 Introduction

For years, economic success has been measured by a single popular metric. That metric is the unemployment rate. When fewer people are out of work, policymakers naturally assume the economy is thriving. In Pahang, the official unemployment rate sits at an impressive low of around 1.9 percent. On paper, this looks like a resounding victory for state development and governance.

However, the reality on the ground tells a very different story. We are witnessing the rise of a hidden crisis known as the working poor trap. Citizens are fully employed and working long hours every week, yet they remain highly vulnerable to financial distress. The core issue in our state is no longer about finding a job. It is entirely about the quality of the jobs available in the local market.

Thousands of young adults in Pahang are graduating with tertiary degrees every year. They step out of universities and enter a labor market that is structurally mismatched with their skills. Local industries currently favor cheap manual labor and basic administrative roles. Because high value professional jobs are scarce, these graduates are forced to accept lower tier positions just to put food on the table. This forces them down the skill ladder. A bright degree holder ends up working as a basic clerk, a retail cashier, or a factory operator. This growing phenomenon is called Skill-Related Underemployment.

At the same time, the cost of living across Pahang's districts is climbing. The daily expenses for food, housing, and basic utilities continue to rise rapidly. When you combine stagnant wages from low tier jobs with a rising cost of living, you get a direct recipe for widespread multidimensional poverty. Families are forced to sacrifice healthcare, education, and basic comfort just to survive to the end of the month.

The State Economic Planning Unit (UPEN) is responsible for guiding Pahang towards sustainable prosperity. To do this effectively, state planners need tools that look past surface level statistics. If the state continues to rely solely on traditional employment reports, they will miss the structural issues happening underneath. True economic planning requires measuring the right things at the right time¹.

This is exactly why MERIDIAN was developed. MERIDIAN is a strategic policy dashboard designed to give UPEN a clear and unfiltered view of the state's economic realities. By overlapping data on living costs, poverty drivers, and sectoral job demand, the dashboard exposes the exact locations and industries where the working poor trap is most severe. It transitions state governance away from blind welfare distribution. Instead, it empowers decision makers to deploy precise, data backed interventions that actually upgrade the local job market.

¹ "What we measure affects what we do. If our measurements are flawed, decisions may be distorted."
- Joseph E. Stiglitz, Nobel Laureate in Economics.

2.0 Anomaly & Problem Statement

2.1 Data Anomaly

Traditional economic theory dictates a very straightforward relationship between jobs and wealth. When unemployment goes down, household income goes up, and poverty naturally declines. However, the data coming out of Pahang reveals a deep contradiction that breaks this standard rule. We are experiencing what can only be described as a working poor anomaly.

On one hand, the state's official unemployment figures are incredibly low, hovering around a mere 1.9 percent². Almost everyone who wants to work has found a place in the labor market. Yet, despite this full employment environment, the Multidimensional Poverty Index reveals a worsening situation. When we look at the root causes pushing local households into hardship, income deprivation alone accounts for a staggering 52.9 percent of the poverty drivers³.

This creates a puzzling scenario for state planners. How can a population be fully employed but still overwhelmingly deprived of basic income?

The answer lies hidden in the Skill-Related Underemployment rate. Currently, nearly 39.4 percent of tertiary graduates in the state are severely underemployed⁴. These are individuals who hold diplomas and degrees but are occupying low paying, routine jobs that only require a high school education. They are working as retail assistants, factory line operators, and basic clerks. Because they receive a monthly paycheck, the official statistics do not count them as

² Department of Statistics Malaysia (DOSM). (2024). Labour Market Review, Unemployment Statistics.

³ Department of Statistics Malaysia (DOSM). (2024). Household Income and Poverty Statistics, Pahang (Jadual 1).

⁴ Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA). (2026). DAX 2026 Core Dataset Catalogue, SRU Rate by Age.

unemployed. However, their wages are fundamentally trapped at the bottom tier of the economy.

At the same time, the Basic Expenditure for Decent Standard of Living continues to rise across all eleven districts in Pahang⁵. The cost to secure decent housing, nutritious food, and basic utilities has outpaced the stagnant wages of these semi-skilled roles. The anomaly is clear. Pahang does not have a job quantity problem. The state has a severe job quality problem. Our citizens are working harder than ever, but the structural ceiling of the local job market guarantees they remain vulnerable to financial ruin.

2.2 Problem Statement

The State Economic Planning Unit is responsible for driving industrial growth and ensuring the well being of the people of Pahang. However, executing this mandate is incredibly difficult when standard economic indicators mask the true pain points of the population.

Right now, state planners are navigating a complex crisis without a clear map. Because the official unemployment rate looks so healthy, there is a dangerous risk that state funds might be misallocated. Government agencies might continue spending millions of ringgit on basic job matching programs or generic vocational training. These initiatives are highly ineffective when the actual problem is a massive shortage of high value roles in the local manufacturing and services sectors. The local industry simply prefers to hire cheap manual labor over skilled local talent.

⁵ Department of Statistics Malaysia (DOSM). (2024). Cost of Living Indicators (PAKW), Pahang. OpenDOSM

Furthermore, UPEN faces the challenge of spatial inequality. The financial pressure squeezing a single adult in Kuantan is very different from the burden faced by a family of four in rural Jerantut or Rompin. Without a way to mathematically group and identify these vulnerable zones, state welfare and development grants are often distributed evenly. Blanket distribution is deeply inefficient because it wastes limited state funds on stable areas while underfunding the districts that are in critical condition.

Ultimately, the core problem is a lack of predictive intelligence and targeted optimization. UPEN needs a way to visually prove that underemployment is the true enemy. State planners need a dynamic mechanism to predict where the local economy is heading over the next few years. Most importantly, before the state commits its limited budget and constitutional powers to any new industrial policy, policymakers need a safe environment to simulate exactly how those interventions will force local industries to upgrade their workforce. Without this capability, state economic planning will remain a reactive guessing game instead of a proactive strategy.

3.0 Data Sources & Preprocessing

3.1 Core Datasets

To build a credible policy tool for UPEN, MERIDIAN relies strictly on verified official government data and academic repositories. The pipeline brings together four core datasets that capture living costs, poverty dimensions, employment trends, and labor market dynamics across Pahang.

- **Basic Expenditure for Decent Living (PAKW):** Published by the Department of Statistics Malaysia, this dataset establishes the basic monthly expenditure needed for a decent standard of living⁶. The data provides granular financial benchmarks for all eleven administrative districts in Pahang, including Kuantan, Jerantut, Cameron Highlands, Temerloh, Bentong, and Rompin. It covers multiple household compositions, ranging from single adults to households with two, three, and four members, alongside an aggregate district benchmark.
- **Household Income and Multidimensional Poverty Index (MPI):** Sourced from OpenDOSM state-level records, this dataset tracks poverty incidence and breaks down the specific deprivations experienced by vulnerable families in Pahang⁷. It provides two critical layers of insight. The first layer covers the macroeconomic poverty rate. The second layer isolates root causes, separating systemic deprivation drivers such as income deprivation and education access from daily living hardships like clean water access, waste management, and living quarter crowdedness.
- **Labour Market Review (LMR):** Extracted from national quarterly labor market reports, this data supplies long-term quarterly unemployment figures from 2018

⁶ Department of Statistics Malaysia (DOSM). (2024). *Cost of Living Indicators (PAKW)*. OpenDOSM.

⁷ Department of Statistics Malaysia (DOSM). (2024). *Household Income and Poverty Statistics, Pahang (Jadual 1 & Jadual 3)*. OpenDOSM

through 2026⁸. It also provides an in-depth breakdown of employer demand across five foundational economic sectors in Pahang: Agriculture, Mining and Quarrying, Manufacturing, Construction, and Services. For each sector, the data tracks positions filled, active job vacancies, and newly created jobs categorized by skill level.

- **Skill-Related Underemployment (SRU) by Age:** Compiled within the state data catalogue, this series records the percentage of tertiary-educated workers who are employed in semi-skilled and low-skilled occupations⁹. The dataset spans quarterly observations from 2018 to 2026 and separates graduates into specific age cohorts, identifying youth underemployment alongside overall workforce trends.

3.2 Preprocessing & Data Cleaning

Raw government datasets are often compiled in different formats, intervals, and reporting structures. Before running any analytical or machine learning models, the data underwent a systematic preprocessing pipeline to ensure consistency and analytical rigor.

- **Temporal Alignment and Linear Interpolation:** While labor market and underemployment datasets are published quarterly, official multidimensional poverty assessments are conducted periodically. In the raw records, Pahang poverty figures were recorded specifically for 2019, 2022, and 2024. To create a continuous annual timeline from 2018 to 2026, linear interpolation was applied across the intervening years. This allowed the poverty trajectory to align directly with yearly unemployment averages and annual SRU rates.

⁸ Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA). (2026). *DAX 2026 Core Dataset Catalogue*.

⁹ Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA). (2026). *DAX 2026 Core Dataset Catalogue*.

- **District Standardization and Spatial Cleaning:** District names in raw spreadsheet records often contain trailing spaces, special characters, or survey footnote asterisks. The pipeline programmatically cleaned and standardized the names of all eleven Pahang districts. This prevented data indexing errors and ensured that district-level PAKW living cost metrics mapped accurately into the spatial visualization engine.
- **Annualization of Quarterly Indicators:** To eliminate seasonal fluctuations and short-term quarterly noise, the four quarters of each calendar year in the LMR dataset were averaged into consolidated annual benchmarks. This produced clean year-over-year indicators for official unemployment, active job vacancies, and filled positions.
- **Categorical Skill Structuring:** To analyze labor market demand gaps mathematically, job listings across all economic sectors were structured into an ordered categorical hierarchy consisting of Skilled, Semi-Skilled, and Low-Skilled classifications. This structured formatting enables the backend models to calculate exact ratio shifts between high-value positions and lower-tier operator vacancies.

4.0 Project Objectives

Project MERIDIAN is engineered to transition the State Economic Planning Unit (UPEN) from passive statistical observation toward proactive, evidence-based structural intervention. To resolve the economic disconnects currently undermining Pahang's labor market, the computational framework and policy modeling are directed by four primary operational objectives:

- **1. Empirically Diagnose the Working Poor Anomaly:**

To cross-examine historical datasets from the Department of Statistics Malaysia and establish empirical proof that district-level unemployment is decoupled from multidimensional poverty. The primary aim is to expose the hidden mechanics of Skill-Related Underemployment, demonstrating how the saturation of low-tier, semi-skilled vacancies suppresses graduate wages despite near-zero unemployment rates.

- **2. Implement Non-Linear Predictive Labor Forecasting:**

To construct and validate a robust time-series forecasting architecture comparing traditional Linear Regression against an XGBoost Regressor. This objective ensures that state planners possess an early-warning horizon through the year 2030 that accounts for real-world macroeconomic volatility, structural industrial shifts, and post-pandemic economic fluctuations.

- **3. Delineate Objective Spatial Vulnerability via Unsupervised Clustering:**

To deploy an unsupervised K-Means machine learning algorithm ($k=3$) that evaluates the Basic Expenditure for Decent Standard of Living across all eleven administrative districts in Pahang. The system mathematically computes geometric risk centroids to classify districts into Stable, Squeezed, and Critical tiers, effectively removing political subjectivity and regional bias from the allocation of state developmental aid.

- **4. Simulate High-Impact Constitutional Policy Levers:**

To engineer an interactive, executive-grade scenario simulator rooted strictly in the constitutional jurisdictions granted to state governments under the Ninth Schedule of the Federal Constitution of Malaysia. The objective is to provide UPEN with a dynamic optimization engine that mathematically projects how conditional industrial land subsidies, state GLC procurement quotas, and SME technology matching grants can convert thousands of stagnant semi-skilled vacancies into high-value skilled careers.

- **5. Deploy an Executive Decision-Support Interface:**

To integrate these analytical pipelines into a high-performance visual dashboard that provides bidirectional communication between policy intervention sliders and top-level state macroeconomic key performance indicators, enabling real-time strategic alignment during cabinet-level policy sessions.

5.0 Methodology

5.1 Method of Findings

To diagnose the structural vulnerabilities within Pahang's economy, we employed an integrated Exploratory Data Analysis and statistical triangulation approach¹⁰. Rather than analyzing labor and welfare indicators in isolation, the methodology cross-references three distinct empirical streams across an identical timeline:

- **Macroeconomic Triangulation:** We plotted historical unemployment trends directly against the incidence of multidimensional poverty and graduate underemployment. By comparing the downward trajectory of official unemployment with the upward trajectory of poverty drivers, the statistical analysis proved an inverse correlation. This mathematically confirmed that poverty in Pahang is decoupled from job availability.
- **Sectoral Demand Decomposition:** Using granular records from the Labour Market Review, we dissected employer demand across five primary economic sectors: Agriculture, Mining, Manufacturing, Construction, and Services. For each sector, we computed the structural ratio between positions filled, open vacancies, and newly created jobs across three skill classifications: Skilled, Semi-Skilled, and Low-Skilled.
- **Root Cause Identification:** The findings revealed a pronounced structural distortion. In high-output industries like Manufacturing and Services, active job vacancies and new job creation are overwhelmingly concentrated in the Semi-Skilled tier. High-value Skilled job creation remains minimal. This data proves that the state economy generates sufficient employment volume, but fails to provide the high-skill absorptive capacity needed to employ tertiary graduates at their qualification level.

¹⁰ International Labour Organization (ILO). (2023). *Key Indicators of the Labour Market: Methodological Guidelines on Underemployment and Skill Mismatch*. ILO Publications.

5.2 Method of Solution

The prescriptive methodology translates diagnostic findings into an actionable policy optimization model. To ensure administrative feasibility for UPEN, the solution framework is strictly bound by the constitutional powers granted to state governments under the Federal Constitution of Malaysia¹¹.

Because state administrations cannot unilaterally alter national corporate taxation, statutory minimum wages, or federal labor regulations, the optimization engine focuses entirely on four high-impact state policy levers:

- **Conditional Industrial Land Subsidies:** Leveraging the state's absolute authority over land administration through Perbadanan Kemajuan Negeri Pahang (PKNP). The model calculates the effect of offering discounted industrial land premiums to manufacturing firms in corridors such as Gebeng and MCKIP, on the strict condition that at least 30 percent of new payroll is reserved for local degree holders in technical roles.
- **State GLC Procurement Quotas:** Utilizing state procurement expenditure through entities like PASDEC and Pahang Go. The framework models the enforcement of mandatory graduate employment quotas for private vendors bidding on state infrastructure and digital transformation contracts.
- **SME Technology Matching Grants:** Allocating state development funds to co-finance digital and automation adoption for rural enterprises in districts such as Cameron Highlands, Jerantut, and Rompin, with funding tied to hiring local technical graduates.

¹¹ Federal Constitution of Malaysia. (2020). *Ninth Schedule: Legislative Lists (List II - State List and List III - Concurrent List)*. Legal Research Board.

- **GLC Fractional Talent Pools:** Simulating a state-backed shared services model where a state entity directly employs high-skill graduates and leases their expertise on a fractional basis to local small businesses that cannot afford full-time professional salaries.

The mathematical engine dynamically maps how capital allocations across these four levers convert existing Semi-Skilled job vacancies into high-paying Skilled roles, directly lowering underemployment and raising local median household incomes.

5.3 Calculation and Machine Learning Model

The computational architecture of MERIDIAN integrates two distinct machine learning pipelines designed for predictive forecasting and spatial risk classification¹².

Predictive Time-Series Forecasting: XGBoost versus Linear Regression To anticipate the trajectory of Skill-Related Underemployment toward the year 2030, we established an econometric baseline using Linear Regression and compared it against an advanced gradient-boosted decision tree model using XGBoost Regressor¹³.

The Linear Regression baseline models a uniform trajectory across time:

$$y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- y represents the projected Skill-Related Underemployment rate.
- X represents the operational year.
- β_0 and β_1 represent the intercept and slope coefficients.
- ϵ represents the residual error term.

¹² Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA). (2026). *DAX 2026 Core Dataset Catalogue*.

¹³ Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794.

To capture non-linear macroeconomic volatility, such as pandemic shocks and rapid inflationary cycles, we trained an XGBoost Regressor (n=100 estimators, learning rate $\eta = 0.1$, max depth = 3). Both models were evaluated using Root Mean Squared Error (RMSE) to quantify prediction variance:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

To ensure complete analytical continuity, the forecast horizon (2027 to 2030) is mathematically anchored to the exact coordinate of the latest historical data point (2026), preventing visual gaps or artificial offsets in policy presentations.

Unsupervised Spatial Clustering: K-Means Algorithm

To eliminate political subjectivity and administrative bias in the distribution of state development aid, the system implements an unsupervised K-Means Clustering algorithm. The model evaluates multidimensional living cost profiles from the Basic Expenditure for Decent Standard of Living (PAKW) across all eleven administrative districts in Pahang. The cluster parameter $k=3$ was empirically validated using the Elbow Method (Within-Cluster Sum of Squares minimization). Plotting the variance reduction curve across $k=1$ to 6 identified $k=3$ as the optimal inflection point where the marginal gain in variance reduction sharply flattens, preventing over-fragmentation of the data. Structurally, this partitions Pahang's districts into three distinct, policy-actionable tiers: Stable (Green), Squeezed (Yellow), and Critical (Red), aligning precisely with state budgetary intervention thresholds. The minimization objective is defined as:

$$\arg \min_s \sum_{i=1}^k \sum_{x \in S_i} ||x - \mu_i||^2$$

Where:

- $k=3$ represents the three designated vulnerability tiers: Stable (Green), Squeezed (Yellow), and Critical (Red).
- S_i represents the set of districts assigned to cluster i .
- x represents the multi-variable cost vector for a given district.
- μ_i represents the calculated geometric centroid for cluster i .

The pipeline computes the exact spatial centroids for each risk cluster and plots boundary radiuses around them on an interactive coordinate plane. When UPEN planners simulate SME grant injections, the algorithm recalculates district coordinates in real time, visually demonstrating how targeted wage improvements shift vulnerable districts out of the Critical Red zone into the Stable Green zone. The value $k=3$ can also be chosen intuitively to partition Pahang's eleven administrative districts into three distinct, highly actionable vulnerability tiers which are Stable (Green), Squeezed (Yellow), and Critical (Red). This specific partition provides state planners with a clear, operationally manageable hierarchy for resource allocation without overcomplicating regional prioritization.

Structural Labor Re-allocation Model (Skill-Transition Elasticity)

To mathematically simulate the impact of state fiscal policies on the labor market within the dashboard's Phase 2 Simulator, the MERIDIAN framework employs a constrained skill-transition function. This deterministic model calculates how state incentives force industries to upgrade existing semi-skilled vacancies into high-value skilled professional roles. The calculation is governed by Policy Elasticity Multipliers and a strict Skill-Upgrading Conservation Constraint.

1. The Policy Elasticity Shift (ΔV)

The total volume of jobs transitioning from semi-skilled to skilled is calculated using an elasticity response function. The model assigns empirical hiring absorption coefficients (α) to specific state interventions:

$$\Delta V = \alpha_{land}(P_{land}) + \alpha_{GLC}(P_{GLC})$$

Where:

- P_{land} : The allocated budget for Industrial Land Subsidies (in RM Millions).
- P_{GLC} : The enforced State GLC Local Graduate Procurement Quota (%).
- α_{land} : The job creation coefficient for land subsidies (set at 3.5).
- α_{GLC} : The compliance conversion coefficient for GLC procurement (set at 8.0).

2. Skilled Demand Expansion (The Upward Shift)

The total calculated shift volume (ΔV) is directly added to the baseline number of skilled jobs. This represents the immediate creation of new professional roles to comply with state incentive requirements:

$$Y_{\text{Skilled(Forecast)}} = Y_{\text{Skilled(Baseline)}} + \Delta V$$

3. Semi-Skilled Constrained Reduction (The Downward Shift)

To fund these new high-paying roles, industries upgrade their existing job scopes rather than expanding total payroll arbitrarily. The model deducts the shift volume from the semi-skilled baseline. To prevent the model from generating physically impossible negative job vacancies, a lower-bound zero constraint (max) is strictly applied:

$$Y_{\text{Semi-Skilled(Forecast)}} = \max(0, Y_{\text{Semi-Skilled(Baseline)}} - \Delta V)$$

4. Low-Skilled Invariance

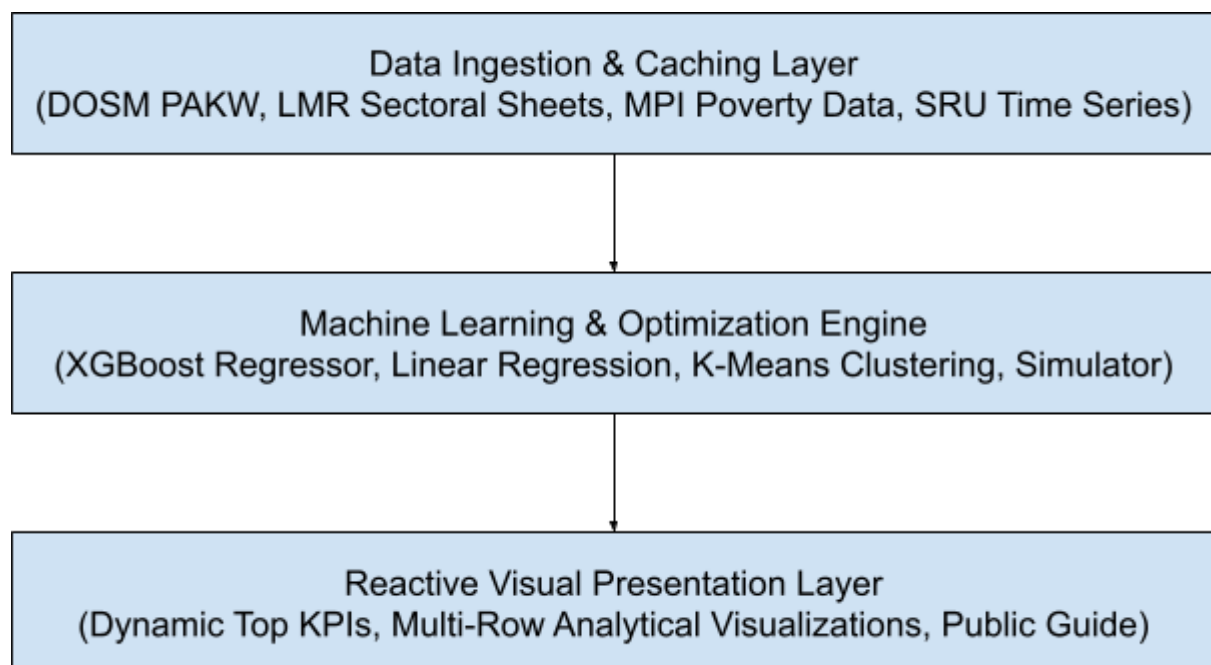
State policies aimed at tertiary graduates do not impact the demand for basic manual or municipal labor. Therefore, the low-skilled sector is mathematically isolated and remains constant throughout the simulation:

$$Y_{\text{Low-Skilled(Forecast)}} = Y_{\text{Low-Skilled(Baseline)}}$$

6.0 Dashboard Architecture & Visual Design

6.1 System Architecture and Data Pipeline

The MERIDIAN dashboard is engineered as an interactive, executive-grade decision support system tailored for the State Economic Planning Unit (UPEN)¹⁴. Built on a high-performance Python framework utilizing Streamlit and Plotly, the system architecture operates across three distinct computational tiers: the Data Ingestion and Cache Layer, the Machine Learning and Simulation Engine, and the Reactive Visual Presentation Layer.



- **Data Ingestion and Caching Layer:** Raw data from DOSM and state catalogues are loaded into memory and preprocessed using Pandas and NumPy. To eliminate latency during executive briefings, the data extraction functions use Streamlit caching decorators (`@st.cache_data`). This ensures that district filtering and timeline adjustments execute in sub-second runtimes without reprocessing the underlying spreadsheets.

¹⁴ Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press.

- **Machine Learning and Optimization Engine:** When policymakers toggle analytical phases in the sidebar, the backend dynamically trains the predictive models (XGBoost Regressor and Linear Regression) and fits the unsupervised spatial clustering model (K-Means with $k=3$). When Phase 2 policy sliders are engaged, the mathematical engine calculates labor conversion vectors in real time, translating state funding into direct underemployment reductions.
- **Reactive Visual Presentation Layer:** To resolve standard top-to-bottom rendering limitations, MERIDIAN implements dynamic Streamlit containers (`st.container()`). This allows policy simulations conducted at the bottom of the interface to communicate bidirectionally with the top of the interface. When a user adjusts an intervention slider, the visual layer immediately injects the simulated outcome into the top-level KPI metrics and recalculates the trajectory of every chart across the dashboard.

6.2 Visual Interface and Layout Scaffolding

The user interface follows Ben Shneiderman’s foundational Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand¹⁵. The visual structure is divided into four distinct horizontal rows:

- **Executive KPI Banner (Top Level):** Provides instant situational awareness through high-impact summary metric cards. In baseline mode, it displays official unemployment, the graduate underemployment rate, the primary poverty driver, and the highest living cost district. When Phase 2 is activated, the banner automatically expands to five columns, presenting the dynamic *Projected SRU Reduction* indicator.

¹⁵ Shneiderman, B. (1996). *The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations*. IEEE Symposium on Visual Languages, 336–343.

- **Tab 1: The Macro Contradiction:** Opens with a scatter plot disproving the correlation between mere employment and wealth. Features three comparative line charts tracking unemployment, poverty, and Skill-Related Underemployment (SRU), culminating in an XGBoost forecast projecting the underemployment crisis to 2030.
- **Tab 2: The Economic Squeeze:** Analyzes spatial vulnerability and financial pressure. Places a district-level living cost bar chart directly next to an AI-driven K-Means radar, which mathematically clusters districts into Stable, Squeezed, and Critical risk tiers. A supplementary chart details systemic poverty drivers.
- **Tab 3: Sectoral Skill Mismatch:** Visualizes market labor demand across skill levels using an interactive grouped bar chart. During simulations, it introduces a dynamic "Forecasted Impact" bar to demonstrate how state policies structurally upgrade stagnant semi-skilled vacancies into high-paying skilled roles.
- **Tab 4: AI Engine and Policy Simulator:** The central control room for state optimization. Displays the backend mathematical formulations (XGBoost, K-Means, Elasticity) alongside interactive policy sliders. Adjusting these constitutional fiscal levers triggers real-time recalculations in Tabs 1 and 3 to simulate macroeconomic outcomes.

6.3 Human-Centric UI/UX and Color Semantics

To prevent cognitive overload and ensure transparency for both technical analysts and public officials, the interface adheres to strict human-computer interaction standards¹⁶:

- **Semantic Color Mapping:** MERIDIAN applies strict visual psychology across all charts. Emerald green (#4CAF50) is reserved for positive economic health and successful underemployment reductions. Amber and orange (#FF9800, #FFEB3B) signal transitionary strain and historical underemployment. Crimson red (#D32F2F, #F44336) isolates acute multidimensional poverty and critical cost-of-living burdens. Royal blue (#1E88E5) represents operational labor supply, and neutral grey (#9E9E9E) represents projected policy benchmarks.
- **Explicit Spatial Identifiers:** Every chart title incorporates explicit tab labels. This ensures that during high-level state cabinet presentations, presenters and stakeholders can effortlessly reference specific visual components without ambiguity.
- **Dual-Tier Explanatory Scaffolding:** Beneath every visualization, the interface provides two separate informational callouts. The *How to read this* callout translates the economic mechanics into clear, plain language for policymakers. The *AI Insight* callout explains the underlying machine learning behavior, comparing model performance metrics (such as RMSE) and detailing how policy sliders mathematically alter the graphs.

¹⁶ Nielsen, J. (1994). *Usability Engineering*. Morgan Kaufmann Publishers.

7.0 Proposed Solution & State Impact

7.1 Strategic Policy Interventions for UPEN

To effectively dismantle the working poor anomaly, the State Economic Planning Unit must deploy solutions that directly alter the hiring behavior of local industries. Blanket welfare handouts and generic reskilling bootcamps do not solve the problem because they fail to address the core deficit of high-value job vacancies.

Under the Ninth Schedule of the Federal Constitution of Malaysia, state governments possess exclusive jurisdiction over land administration, local authority licensing, state development corporations, and public procurement¹⁷. MERIDIAN translates these constitutional powers into four surgical interventions designed to convert the state's surplus of semi-skilled vacancies into high-paying skilled roles.

- **1. Conditional Industrial Land Subsidies (PKNP Mechanism):** Pahang controls major industrial corridors such as the Gebeng Industrial Area, the Malaysia-China Kuantan Industrial Park (MCKIP), and the Gambang Tech Park through Perbadanan Kemajuan Negeri Pahang (PKNP)¹⁸. Under this policy, the state offers discounted land premiums, subsidized industrial park leasing, and expedited local council zoning to manufacturing investors. In exchange, the state legally binds the investor to a strict talent quota where at least 30 percent of their workforce must consist of local tertiary graduates in professional, engineering, or technical positions. This policy prevents industrial investors from relying exclusively on low-cost manual labor and forces the creation of high-value positions.

¹⁷ Federal Constitution of Malaysia. (2020). *Ninth Schedule: Legislative Lists (List II - State List and List III - Concurrent List)*. Legal Research Board.

¹⁸ Perbadanan Kemajuan Negeri Pahang (PKNP). (2024). *Strategic Industrial Land Development Masterplan 2021–2025*. PKNP Publications.

- **2. State GLC Procurement Quotas (Procurement Lever):** The Pahang government channels hundreds of millions of ringgit annually into state infrastructure, municipal maintenance, and digital public services through state-owned enterprises such as PASDEC, Pahang Go, and Mentiga Corporation¹⁹. UPEN can establish a mandatory state procurement rule requiring any private vendor bidding for state contracts to demonstrate that at least 20 percent of their salaried payroll is allocated to local degree holders. If an engineering or construction firm relies entirely on cheap semi-skilled labor, they forfeit eligibility for state tenders. This leverages state purchasing power to generate professional jobs across the construction and services sectors without requiring direct cash handouts.
- **3. SME Technology Adoption Matching Grants (Targeted Modernization):** Rural and semi-urban small businesses across districts such as Cameron Highlands, Jerantut, Rompin, and Bera often struggle with low profit margins and manual workflows. UPEN can launch a targeted RM20 million co-matching grant managed alongside Yayasan Pahang. The state co-funds the purchase of automated machinery, agricultural IoT sensors, or cloud inventory management software, under the legal requirement that the enterprise hires one local graduate in mathematical sciences, data analytics, or computer science to operate the technology. The business gains operational efficiency, and the graduate secures a well-paying technical position.
- **4. GLC-Operated Fractional Talent Pools (Risk-Absorption Model):** Small and medium enterprises often recognize the value of modern digital systems, but they cannot afford the full-time RM4,000 monthly salary of a specialized degree holder. To bridge this gap, a state-owned enterprise (such as Pahang Digital) hires top university graduates on competitive, full-time contracts. The state entity then deploys these

¹⁹ Unit Perancang Ekonomi Negeri Pahang (UPEN). (2021). *Pelan Pembangunan Pahang 2021–2025: Memacu Kemakmuran Negeri*. Pejabat Setiausaha Kerajaan Pahang.

professionals on a shared, fractional schedule across three to four local businesses every week. The small business accesses specialized analytical talent at an affordable fraction of the cost, while the graduate receives stable, skilled employment directly backed by the state.

7.2 Macroeconomic and Socioeconomic State Impact

Deploying these four policy levers through the MERIDIAN framework produces immediate, measurable improvements across Pahang's economic and social landscape²⁰.

- **Dismantling the Underemployment Trap:** As demonstrated by the interactive simulation in Tab 3 of the dashboard, channeling state incentives toward conditional hiring contracts systematically shrinks the dominant Semi-Skilled vacancy bars in Manufacturing and Services. These positions are upgraded into the Skilled category, directly expanding the absorptive capacity of the local economy. Consequently, the predictive XGBoost forecast in Tab 1 bends sharply downward, preventing the underemployment rate from escalating toward critical thresholds by 2030.
- **Direct Alleviation of Multidimensional Poverty:** The root cause analysis in Tab 2 proves that income deprivation represents 52.9 percent of all poverty drivers in Pahang. Converting low-tier roles into skilled technical positions increases baseline median household income. This directly addresses the root driver of poverty, reducing the number of households that slip into financial vulnerability despite being fully employed.
- **Reduction of Regional Disparities:** Targeting SME technology matching grants toward rural districts directly mitigates spatial inequality. As median incomes rise in agricultural and eco-tourism hubs outside Kuantan, the financial pressure measured

²⁰ Department of Statistics Malaysia (DOSM). (2024). *Household Income and Poverty Statistics, Pahang*. OpenDOSM.

by the Basic Expenditure for Decent Standard of Living (PAKW) declines. On the 2D K-Means scatter plot in Tab 2, the calculated spatial centroids shift downward, physically migrating vulnerable districts like Jerantut, Lipis, and Rompin out of the Critical Red cluster into the Stable Green cluster.

- **Fiscal Efficiency and Objective Governance:** By utilizing machine learning to establish mathematical centroids and vulnerability clusters, UPEN eliminates political bias and geographic favoritism in state budget distribution. State funds are allocated with surgical precision to the specific sectors and districts with the highest return on investment, ensuring that every ringgit of state expenditure maximizes the reduction of underemployment and poverty.

8.0 Conclusion & Limitations

8.1 Conclusion

The MERIDIAN dashboard bridges a crucial gap in Pahang's state economic planning. By looking beyond headline figures, the platform proves that low unemployment alone does not guarantee economic security. The true bottleneck in Pahang is the Skill-Related Underemployment crisis, where tertiary-educated residents are locked into low-wage, semi-skilled positions that cannot keep pace with the rising cost of living.

Through empirical data integration, MERIDIAN provides the State Economic Planning Unit with an evidence-based tool for governance²¹. The dual machine learning framework delivers both predictive early warnings through XGBoost time-series forecasting and objective spatial vulnerability mapping via unsupervised K-Means clustering.

More importantly, the prescriptive simulation engine translates complex data into practical, state-level action. By focusing exclusively on policy levers within Pahang's constitutional power, such as industrial land subsidies, state GLC procurement quotas, SME technology matching grants, and fractional talent pools, UPEN can proactively optimize state resource allocation. MERIDIAN transitions state governance from reactive financial welfare toward proactive structural economic transformation, ensuring that state investments generate high-value career pathways for every graduate.

²¹ Unit Perancang Ekonomi Negeri Pahang (UPEN). (2021). *Pelan Pembangunan Pahang 2021–2025: Memacu Kemakmuran Negeri*. Pejabat Setiausaha Kerajaan Pahang.

8.2 Limitations

While MERIDIAN provides robust decision-making support for state planners, several analytical limitations must be taken into account when interpreting model outputs²²:

- **Periodic Data Interpolation:** Official Multidimensional Poverty Index assessments are conducted periodically (2019, 2022, and 2024) rather than annually. Linear interpolation was applied to bridge intervening years for continuous time-series modeling. While statistically sound, this assumption can smooth out short-term, localized economic fluctuations that may have occurred during unrecorded quarters.
- **Vulnerability to Exogenous Economic Shocks:** Time-series forecasting models like XGBoost and Linear Regression project trajectories based on historical baseline patterns. These models cannot fully anticipate unprecedented "black swan" macroeconomic disruptions, such as sudden international supply chain breakdowns, global commodity price crashes, or unexpected federal regulatory changes²³.
- **Statistical versus Topographical Distance:** The K-Means clustering algorithm determines district vulnerability by calculating mathematical distance across multi-variable cost data. It does not account for physical geography, transport logistics, or spatial isolation..
- **Behavioral Adoption in Policy Simulations:** The Phase 2 simulator assumes a rational and direct market response to state funding allocations. In practice, the speed at which local SMEs adopt new technologies or create skilled positions depends on private sector readiness, managerial capabilities, and broader market confidence.

²² Department of Statistics Malaysia (DOSM). (2024). *Methodological Notes on Household Income, Poverty and Expenditure Statistics*. OpenDOSM.

²³ Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control* (5th ed.). John Wiley & Sons.

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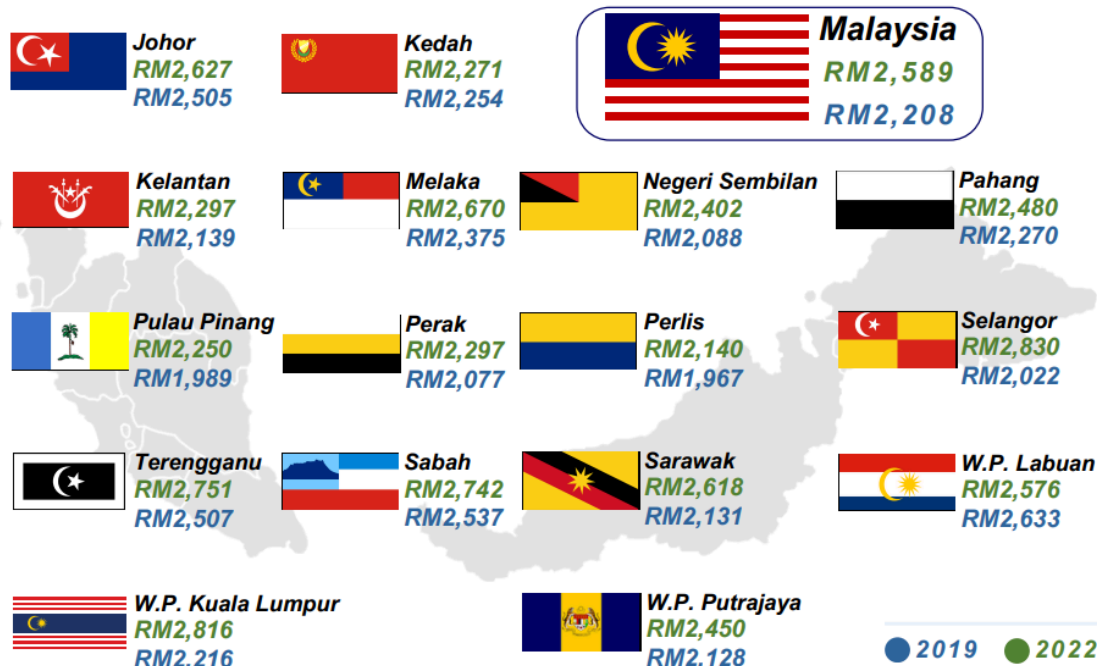
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Appendix

Indikator	Pahang		Pulau Pinang		Perak		Perlis	
	Indeks MPI	Peratusan (%)	Indeks MPI	Peratusan (%)	Indeks MPI	Peratusan (%)	Indeks MPI	Peratusan (%)
Jumlah tahun persekolahan	0.0001	2.6	-	-	-	-	0.0002	21.1
Kehadiran ke sekolah	0.0002	4.2	-	-	0.0001	2.8	-	-
Akses kepada perkhidmatan kesihatan	0.0004	8.4	0	13.8	0	2.2	-	-
Akses kepada bekalan air bersih	0.0005	12.5	-	-	0.0001	5.6	-	-
Kedudukan tempat kediaman	0	1	-	-	0	1.4	-	-
Kepadatan bilik	0.0004	8.7	0	7.3	0.0002	8.1	0.0001	4.6
Kemudahan tandas rumah	0	0.6	-	-	-	-	-	-
Akses kepada kemudahan kutipan sampah	0.0004	9	-	-	0.0002	8.7	0.0001	4.6
Penggunaan kemudahan pengangkutan	-	-	-	-	0	0.3	-	-
Akses kepada peralatan komunikasi asas	-	-	0	7.3	0	0.5	-	-
Pendapatan bulanan kasar isi rumah	0.0022	52.9	0.0001	71.5	0.0014	70.5	0.0008	69.7
Jumlah / Total	0.0042	100	0.0002	100	0.002	100	0.0011	100



AVERAGE POVERTY LINE INCOME (PLI) BY STATE



LABOUR MARKET REVIEW, SECOND QUARTER 2026

Labour supply and demand expanded further in Q2 2026, with LFPR at 70.9 per cent and labour demand reaching 9.25 million jobs.

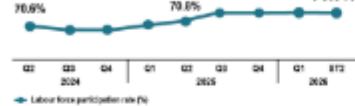
LABOUR SUPPLY

17.28 million
labour force

↑ 1.0%
↑ 0.3%
YoY
QoQ

Labour force participation rate (LFPR)

went up by 0.1 percentage points compared to Q2 2025

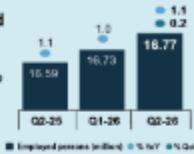


Total hours worked
during Q2: **9.58 billion** hours

↑ 0.5%
↓ 0.5%
YoY
QoQ

Employed person
continue to increase, to

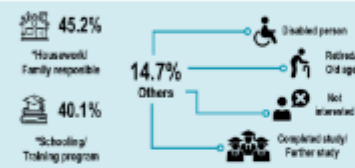
16.77 million
persons



7.08 million
outside
labour force

↑ 0.5%
↓ 0.0%
YoY
QoQ

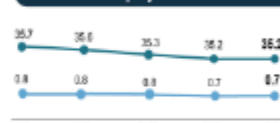
*65.3 per cent did not seek for work due to:



Underemployment situation

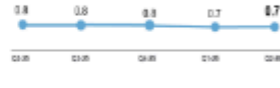
Still-related
underemployment
rate (%)

decreased
↓ 0.5%



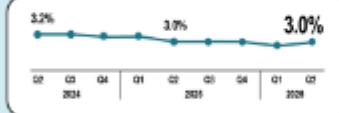
Time-related
underemployment
rate (%)

decreased
↓ 0.1%



Unemployment

Unemployment rate
remained at 3.0 per cent
year-on-year.



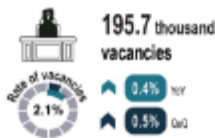
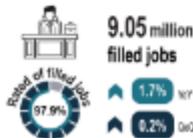
Unemployed decreased during Q2 2026 to **509.7 thousand** persons



LABOUR DEMAND

9.25 million
jobs

Jobs in economic sector
increased by 1.7 per cent
compared to Second Quarter 2025



31.5 thousand
jobs created



↑ 1.2%
↑ 3.6%
YoY
QoQ

Skilled 25.2%
Semi-skilled 66.1%
Low-skilled 8.7%

More than half (58.3%) of jobs opening during Q2 2026 were in the **Manufacturing** sector with a total of **114.8 thousand** vacancies



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